

Alexander Bloomer
424-327-7445 | bloomer@calpoly.edu

EDUCATION

California Polytechnic State University, San Luis Obispo, CA
Bachelor of Science, Computer Science | Expected June 2028

GPA: 4.0

Relevant Coursework: CSC 203: Project-Based Object-Oriented Programming and Design (Java), CPE 225: Introduction to Computer Organization (C, Assembly), Calculus 3, Linear Analysis

El Camino College, Torrance, CA

No Degree

GPA: 3.75

Relevant Coursework: Computer Science 1: Problem Solving and Program Design using C++, Computer Science 2: Introduction to Data Structures (C++)

Redondo Union High School, Redondo Beach, CA

High School Diploma | June 2024

GPA: 4.5

Relevant Coursework: AP Computer Science A, AP Computer Science Principles, AP Calculus AB, AP Calculus BC, AP Physics 1

EXPERIENCE

Boeing

Software Engineering Intern, June 2023 - August 2023

- Developed a robust pipeline for ensuring safety and security in code using Docker containers
- Collaborated with Boeing software engineers to address complex pipeline challenges
- Presented pipeline to employees and chief engineers
- Collaborated with three other interns to present innovative AI strategy to chief engineers and won an AI innovation award

Cal Poly Robotics Club

Member, September 2024 - Present

- Design and develop Django-based website to remotely control a robotic boat
- Collaborate with small team to optimize functionality and usability
- Use Agile development process to enhance teamwork and improve efficiency

Beach Cities Robotics, FIRST Robotics Competition

Lead Technician, August 2021 - August 2022

Programming Student, August 2020 - June 2024

Co-Driver, August 2022 - June 2024

- Collaborated with team members to design, build, and program several robots
- Applied programming concepts to control robots using Java
- Led setup and troubleshooting of the robot as lead technician

- Cooperated with another driver to operate the robot in competitions, winning a regional competition, a world championships division championship, and reaching world championship finals
- Won Excellence in Engineering Award and Entrepreneurship Award

The Coder School

Code Coach, March 2023 - January 2024

- Taught students from 6 to 18 to code in various languages and frameworks including Python, JavaScript, C#, Java, C++, CSS, HTML, Scratch, and Unity
- Engaged students with interesting programming projects to foster creativity and problem-solving skills
- Led summer and winter camps of up to 15 students on topics like Scratch, robotics, and animation

CyberPatriot

Team Member, August 2021 - June 2024

- Collaborated with a 5-person team to secure computer systems by mitigating security threats
- Developed automation scripts to expedite the process and improve the efficiency of risk detection
- Ranked top 20% in the nation and first in school in national competitions

PERSONAL PROJECTS

Baseball Simulator

- Developed a Python-based baseball simulator with Flask web interface
- Processed game outcomes using CSV data with real player statistics from Baseball Reference
- Implemented probability-based calculations for batting outcomes based on players' stats (single, walk, out, etc.) and managerial decisions (e.g., pulling a pitcher)
- Designed understandable and interactive UI using HTML, CSS, and JavaScript, allowing users to easily simulate games or seasons and view results
- Generated detailed game and season results for team analysis
- Published site using Render
- <https://baseball-simulator.onrender.com/> (May take about a minute to load due to not constantly running)

Personal Website

- Designed and built a multi-page personal website using HTML and CSS
- Used best programming practices for version control with Git and GitHub
- <https://alexbloomer.github.io/>

Chrome Extension for Video Speed Control

- Created a Chrome extension that allows users to speed up or slow down videos with custom button clicks
- Implemented JavaScript to modify video playback speed dynamically

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, C#, C, RISC-V Assembly

Software Tools and Frameworks: Unity, Microsoft Excel, Git, GitHub, VScode, IntelliJ, Docker, Bash

Web Development: Django, Flask, HTML, CSS, JavaScript

HONORS

Ivy Chain Award – Top 50 juniors by GPA (2023)

AP Scholar With Distinction (2023)

California Scholarship Federation (2020-2023)

Robotics Programming Student of the Year (2023)

Robotics Rookie of the Year (2021)

Dean's List Honors (2024)